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Predicting CYP inhibition: method report

OpenADMET CYP Inhibition Blind Challenge, direct-inhibition (regression) and time-dependent-inhibition (classification) tracks. Username `GiantWizard`.

What was submitted

The regression entry is a blend of five models over 4,905 training compounds, scoring each of the 750 blind compounds against four cytochrome P450 isoforms. They come from two families: TabPFN regressors over frozen molecular representations, including a pretrained Chemprop embedding and the organisers' `surrogate_adme` encoder, and message-passing networks trained here with thirteen or more prediction heads covering the challenge's own labels, the organisers' single-concentration screen, PubChem potency panels and a ChEMBL extract.

Blend weights are fitted per isoform on development folds. Predictions are then placed onto the blind population's estimated mean and spread, with the scale held at the Murphy optimum for squared error and the centre offset fitted on development rows. For CYP2D6 the weight of the arm trained on pseudo-labelled rows (the first finding below) is set by hand to 0.7, because weights fitted on hit-only rows cannot value it.

Historical leaderboard snapshot on 23 October: **0.4743 macro soft-threshold relative absolute error, rank 41** (upload of 23 October, 23:02 UTC), improved from 0.5471 on the first submission. The best measured upload since is v32 at 0.4669, rank 39 (26 October, 07:53 UTC), and the latest, v34, scored 0.4685 at rank 41 (27 October, 02:57 UTC). The live board scores half of the blind set and the final ranking uses all of it, so each board figure here is a reading on that half.

What this entry offers the Innovation in ML Award

The organisers describe the award as rewarding exploration over leaderboard tuning, judged on architectural novelty, creative data usage, novel uncertainty quantification and clever engineering, independent of rank. This entry does not claim architectural novelty, since every network in it is

a stock Chemprop or TabPFN. The contributions that fit the other three criteria are these, each with the measurement behind it in the sections below.

Uncertainty quantification. The scoring function forgives any prediction that falls inside a compound's credible interval. Treating the interval as a training target, and then as a placement target, shows that interval-aware training and affine calibration are the same intervention reached two ways: each helps alone and they do not stack (0.7442 to 0.7244 before calibration, reversing to 0.7060 against 0.7198 after). A route through a measured response variable failed identically.

Creative data usage. The organisers confirmed that a blank training cell means the compound was not followed up because its primary screen showed no inhibition, and that the test set is dense. Checked against the screen data, every labelled CYP2D6 compound responds at -0.59 or stronger, so a model trained on the labelled rows never sees a weak compound; adding the blank cells as low-valued training rows moved the leaderboard from rank 55 to 41 (first finding below). The organisers built the blind set by purchasing commercial analogues of the strongest training hits. Measuring that structure (median Tanimoto 0.578 to the training set against 0.435 for a scaffold-disjoint holdout) overturned the standard cluster-disjoint validation advice for this challenge. Separately, the five published leaderboard metrics for any entry constrain the blind population's mean and spread, which allowed predictions to be placed without a blind label ever being seen.

Clever engineering. Removing a CPU-era habit of compressing 605 columns to 200 principal components before TabPFN was worth more than any calibration or blending change, and was the only change that improved all five published blind metrics together.

Evidence of exploration. Many directions were run to a measured conclusion, including the ones that failed (see below), and the leaderboard itself was analysed: across 92 ranked entries, soft-threshold error is 92% explained by MAE, R-squared and Spearman, which shows placement could not have closed the gap.

The findings

A model trained only on follow-up compounds has never seen a weak one

The training file has blank pIC50 cells. The organisers confirmed in the challenge Discord on 8 October that a blank means the compound was not selected for dose-response follow-up on that isoform because its primary screen showed no inhibition, and that the test set was built densely, with every compound measured on every isoform.

For CYP2D6 the training data show this sharply. All 1,493 labelled compounds respond at -0.59 or stronger in the organisers' single-concentration screen (log2 fold change), and of the 2,882 blank compounds that have screen data only 5 respond at -0.73 or stronger. A model fitted to the labelled rows alone never meets a weakly active compound, and the blind set holds many. CYP3A4 is different: its blanks include compounds that respond more strongly than the labelled ones, and CYP2C9 overlaps, so the same reading does not apply to them.

The change was to add the blank CYP2D6 cells with a clearly weak screen response (weaker than the 95th percentile of the labelled compounds, -0.72) as training rows at pIC50 3.5, at most one per labelled row. The screen response only selects the rows. It is not an input, since the blind compounds have no such measurement. The model is a TabPFN regressor over the `surrogate_adme` embedding and is otherwise unchanged.

The locked holdout cannot judge this change, because it contains only compounds that were hits,

and it scored the pseudo-labelled arm worse on CYP2D6 (soft-threshold error 0.7966 to 0.8156). Two checks were built for the purpose. For compounds held out of training, the probability that a labelled cell is predicted above a blank one rose from 0.752 to 0.829 across the development folds and from 0.741 to 0.824 on the holdout, while the Spearman correlation on labelled compounds stayed level (0.4460 and 0.4499 on development, 0.5563 and 0.5444 on the holdout). On the 750 blind compounds the share of CYP2D6 predictions below 3.8 went from 1% to 36%, and the ranking changed to a Spearman of 0.72 against the arm without the extra rows.

The leaderboard was the test. The upload with only the CYP2D6 column changed (`v26`) improved all five published metrics at once: macro soft-threshold error 0.4938 to 0.4806, Spearman 0.7234 to 0.7382, R-squared 0.5660 to 0.5845, and rank 55 to 43. The other three columns were identical files, and each macro metric is the mean of the four isoforms, so the CYP2D6 column alone improved by about 0.053 in soft-threshold error and 0.059 in Spearman. A second upload (`v27`) raised that arm's weight in the CYP2D6 blend from 0.4 to 0.7, with the arm averaged over four settings of the floor, the row count and the embedding, and reached 0.4743 and rank 41. The same idea on CYP1A2 changed almost nothing, since the unmodified model already separated blank from labelled cells there at 0.86, and it was not uploaded. The leaderboard scores half of the test set, and each step above is a single upload.

Compressing features destroys an in-context model's advantage

Every TabPFN arm in this project originally reduced its input to 200 principal components before fitting, a habit carried over from working on CPU where the full width would not run. Removing that compression and passing the 605 raw columns directly was worth **+0.0125 of macro development Spearman**, far more than any calibration or blending change made over six weeks.

It was confirmed on blind data rather than only in cross-validation. The submission carrying the change improved on all five published metrics at once: soft-threshold error 0.5433 to 0.5333, mean absolute error 0.7181 to 0.7094, R-squared 0.5211 to 0.5317, Spearman 0.6975 to 0.7041, Kendall 0.5224 to 0.5269. Every previous gain in the project had traded one metric against another, which is the signature of moving predictions rather than improving them.

The memory-saving flags that forced the compression in the first place turn out to cost almost nothing on their own, +0.0023. The damage was entirely in projecting 605 columns onto 200 components chosen to explain variance in the embedding, which is not the same as explaining variance in the target.

Ensemble diversity comes from the pretrained encoder, not from training variation

Four message-passing networks were trained from the same public Chemprop checkpoint, differing in the number of auxiliary prediction heads (thirteen against twenty), the number of random seeds averaged (one against three), whether a ChEMBL extract was included, and whether time-dependent-inhibition heads were attached. All four correlate **above 0.95** with each other on held-out predictions, and once any one of them is in the blend the others contribute nothing. One of them is measurably the best single network of the four and makes the blend worse.

A fifth network, trained identically but starting from a different public checkpoint, correlates **0.70 to 0.90** with the others. It is 0.046 worse than the best one as a standalone model and earns weight 0.25 on three of four isoforms.

So the useful question about a candidate model concerns where its representation came from, more

than how good it is on its own. Seeds, heads, auxiliary tasks and training schedules all produce near-duplicates, and a different pretraining checkpoint produces a different opinion. Later notebooks tested this against ten further encoders and found that the checkpoint already in use outranked all of them (see the encoder screen below).

Cluster-disjoint validation is wrong when the test set is built by analogue expansion

Standard practice in molecular machine learning is to split folds so that chemical clusters never straddle the boundary, on the grounds that random splits flatter a model by letting it memorise series. A competitor ranked well above this entry names exactly that as the change that made their held-out numbers track the leaderboard.

It is the wrong direction here, and the measurement says so plainly. The organisers built the blind set by purchasing commercial analogues of the strongest training hits. Blind compounds therefore sit at **median Tanimoto 0.578** from the training set, with 97% above 0.4 and 40% above 0.6. A scaffold-disjoint holdout sits at **0.435**, with 65% and 5.5%. The holdout is already harder than the thing being scored, which is why holdout error reads near 0.60 while the leaderboard returns 0.51. Filtering it further toward cluster-disjointness moves it away from the regime that matters.

The corollary is that general methodological advice has to be checked against how a particular challenge was constructed, and that the construction is often documented in plain language in the challenge description.

The blind population can be recovered from published leaderboard metrics

The leaderboard reports soft-threshold error, mean absolute error, R-squared, Spearman and Kendall for every entry. For a given submission these are four observable quantities constrained by three unknowns describing the blind population: its mean, its spread, and the entry's correlation with it. Solving that system per isoform recovers those unknowns well enough to place predictions without ever seeing a blind label.

Applied to a competitor's published figures, the inversion disagreed with their stated CYP2D6 blind mean of 3.107 and put it between 3.37 and 3.45. A separate entry subsequently shipped 3.47 and outscored them. The method is not reliable on every isoform; it depends on a normal approximation that the pile-up of inactive compounds violates, and it disagreed with the published Spearman for the other three.

What did not work

Many directions were closed, each with a measurement rather than an impression. The ones worth recording:

Public potency data as features. Models trained on PubChem AID 1851 and the Veith panels rank our own labels at 0.28, 0.34, 0.13 and 0.48 against our models' 0.58, 0.71, 0.43 and 0.81. Strictly worse on every endpoint, and worst on the one they were built to rescue. On CYP2D6 the best public predictor of our labels is a model of a different isoform, which suggests the signal being carried is generic promiscuity.

Scaling the pretraining corpus. A corpus of roughly 150,000 molecules across 667 assay columns, assembled from ToxCast, MUV, HIV, Tox21, PubChem and ChEMBL, produced a worse encoder

than one trained on 33,000 molecules across 20 columns. The large corpus is dominated by binary toxicity assays whose relationship to potency is weak.

Training against the metric directly. The scoring function forgives any prediction inside a compound's credible interval, so refitting each model by majorise-minimise against the clipped prediction should help, and does: 0.7442 to 0.7244 on held-out error for one arm, with the sign consistent across isoforms. The entire effect disappears once the affine calibration stage is applied, reversing to 0.7060 against 0.7198, because both interventions push predictions toward the same credible intervals and they do not stack. An earlier attempt to route predictions through a measured response variable failed in the identical way, which took two independent discoveries to recognise as one phenomenon.

Delta learning. Predicting differences between similar molecules and anchoring on a measured neighbour should suit a test set built by analogue expansion. It reaches 0.4488 and 0.5426 against existing arms at 0.5881 and 0.7060. The reason is an asymmetry that was not checked in advance: blind compounds are very similar to training compounds, but training compounds are not very similar to each other, so pairs have to be drawn at a similarity floor of 0.25 while predictions are made at 0.578.

Changing the multitask network, five ways. Each change was one variable against the same network (`mtl13`, dev macro Spearman 0.6262, holdout 0.6652) with a preset gate of +0.015 dev. Heads on the maximum response, which exists for every row where the fitted potency exists for only 35% of CYP2D6 rows, gave +0.0120 dev with CYP2D6 flat and holdout 0.0099 lower. Swapping the warm start to a different pretrained encoder cost 0.0396 dev. A stock Chemprop recipe (MAE loss, early stopping, fold averaging, ChEMBL heads) scored 0.6030 dev and entered the blend with a paired-bootstrap probability of improvement of 0.12. The organisers' own TDI-condition pIC50 as four extra heads scored 0.6245 dev and 0.6490 holdout, which ties and trails the control. The last of these was predicted in advance, because those labels correlate 0.91 to 0.99 with the direct pIC50.

Ten further molecular encoders into TabPFN. A label-free screen (200 principal components plus the raw screen columns, ridge, scaffold folds) ranked the encoder already in use first at 0.6118 macro, with the best challenger, MiniMol, at 0.6073 and the others from 0.5904 down to 0.5785. The top three then ran through the full recipe and scored 0.6184, 0.6102 and 0.6036 dev against 0.6370 for the incumbent, with the loss concentrated on CYP2C9. Uni-Mol embeddings earlier scored 0.5911 and added nothing to the blend. As blend members the TDI network and MiniMol arms were each worse than the shipped blend on holdout, by 0.0019 to 0.0044, with probabilities of improvement between 0.11 and 0.14.

Pretrained chemical language models as frozen features, both ChemBERTa and MoLFormer, at +0.0014 and +0.0011 macro. **Gradient boosting, random forests, support vector machines and ridge on descriptors**, all 0.05 to 0.08 behind the blend after tuning. **Caruana bagged ensemble selection** over nine arms, which lost to a three-arm grid search. **Mitra**, a tabular foundation model reported to beat TabPFN, which exhausts a T4's memory and falls back to 256 training examples.

Later experiments: 24 to 27 October 2026

MA-ST-RAE is the leaderboard error and is distinct from MAE.

NB33 and the v33 candidate. NB33 fine-tunes the pretrained CheMeleon message-passing

weights at 2048 hidden units and depth 4, with an interval hinge on the four scored heads and 18 auxiliary heads, following the public Scigantic recipe on this project's audited table and scaffold folds. Only CYP2D6 changes: 40% NB33 mixed with 60% incumbent raw predictions, then placed at the incumbent mean and spread. The fixed candidate improved development ST-RAE by 0.008880 across 4/5 folds. On 220 held-out CYP2D6 labels it improved ST-RAE 0.831213 to 0.814041 and Spearman 0.54425 to 0.55725. Removing any one NB33 fold model preserved the gain. The paired scaffold-bootstrap 95% interval for the error difference is [-0.035570,+0.000300], which includes no improvement, and the holdout has been reused. The evidence supported a small incremental candidate.

The rough forecast was 0.463, which was 0.4669 minus the local four-target holdout improvement of 0.004293. That forecast assumed the holdout difference transfers one to one, and the evidence for it was weak.

Additional test	Measured result	Decision
CYP2D6 potency-only and related-assay specialists, five folds each	Best addition to v32 reduced endpoint development error only 0.000446, 3/5 folds; neither improves the fixed NB33 candidate	Exclude
NB29 control versus extra TDI labels, five paired small-model folds	C1 gain only 0.000704, 3/5 folds; C9/C3 best weight zero	No full-size run justified by this pilot
NB33 C1 at 10%, overlooked by best-before-consistency selection	Development improved 5/5 folds; fixed holdout worsened 0.745284 to 0.746917	Exclude
C1 saved-model complement, selected from 38 development comparisons	Fixed holdout worsened 0.745284 to 0.752635; rho also declined	Exclude, despite development gain
C1 potency-category prediction versus matched regression	Best interval-action development gain 0.002542, only 2/5 folds	Exclude

The original NB33 breakout gate remains failed. The incremental checks were separately frozen and authorized; they do not retroactively turn that gate into a pass. The selection-order finding was a limitation of the documented rule, not code contradicting the protocol. Local training peaks were about 0.52GB for specialists and 0.80GB for the TDI pilot.

The v34 upload (v33 with spread rescaling on CYP1A2 and CYP3A4 and a CYP2C9 blend) scored **0.4685, rank 41**, worse than v32, so v32 remains the best uploaded file. Its per-isoform errors were 0.5184 for CYP1A2, 0.3767 for CYP2C9, 0.5728 for CYP2D6 and 0.4061 for CYP3A4. CLAMP embeddings were not evaluated.

Tox21 CYP1A2 as an external predictor (AID 1671199). The assay holds 7,671 unique compounds, and the luciferase counter-screen (AID 1224835) flagged 864 rows, which were removed. A gradient-boosting model trained on 4,995 of the remaining compounds reached Spearman 0.411 on this project's CYP1A2 labels against 0.564 for the incumbent arm, correlated 0.052 with that arm's residual, and moved holdout Spearman from 0.5884 to 0.5905. Only 78 labelled challenge compounds overlap the assay, and the two assays agree on them at Spearman 0.642.

Placement inside the credible interval. The scoring function charges nothing for a prediction inside a compound's credible interval. In the training data the weak CYP2D6 compounds (pIC50

below 3.6) have a median upper bound of 3.75, while the CYP2D6 column in v34 has a median of 3.20 with 65% of its values below 3.5. An offline test used the labelled development compounds with their real intervals plus blank cells whose screen showed no inhibition, given an assumed window from 1.06 to a variable upper bound (64% of that set is inactive, close to the share expected in the blind set). Pulling values below 3.5 toward 3.5 while keeping their order moved the error from 0.557 to 0.517 when the window top is 3.75 and from 0.591 to 0.550 when it is 3.52, and left it at 0.643 when the window top is 3.3. The unmodified column scored 0.557 in this setup against 0.5728 on the board. The assumed values for inactive compounds make this a simulation, and the file carrying the change had not been scored when this was written.

What public write-ups report. Scigantic reports macro error 0.4436 at rank 11 on the interim board from twelve members, eight CheMeleon-based and four Chemprop-based, combined with per-endpoint weights fitted on out-of-fold rows and rescaled to the incumbent's mean and spread (repository). The Workbench entry trains four stock Chemprop networks on 31,670 compounds from the challenge, ChEMBL and the Veith qHTS data and recalibrates each isoform with an affine map (write-up). On the live board of 27 October the leading per-isoform errors were 0.3659 for CYP1A2, 0.2776 for CYP2C9, 0.3952 for CYP2D6 and 0.3162 for CYP3A4. This entry's CYP2D6 column matches entries scoring 0.47 to 0.48 on MAE, R-squared and Spearman and scores 0.573.

Detailed measurements, provenance and restart notes are in the dated continuation record in the repository notes. The method narrative above is the 23 October snapshot; its version-specific descriptions are not the current candidate recipe.

Validation, and where it misled us

A locked holdout of 726 compounds, scaffold-disjoint from development, was scored once per candidate. Across three submissions where the holdout and the leaderboard can be compared, the offset between them was -0.0834, -0.0728 and -0.0833, so two of three agreed to within 0.0001 and predictions from the third submission onward were accurate to 0.0004.

The exception matters. One change improved the holdout by 0.0046 with a paired bootstrap putting the probability of improvement at 0.90, and then scored 0.0060 worse on the blind set. The pattern that has held since is that confidence intervals excluding zero transfer and intervals crossing zero do not, which is a small sample and an uncomfortable rule to rely on. A 726-compound holdout cannot resolve effects of 0.005, and several decisions in this project were made as though it could.

A second exception runs the other way and is larger. The holdout contains only hit compounds, and it scored the pseudo-labelled CYP2D6 arm 0.019 worse and the higher-weight blend a further 0.021 worse, while the leaderboard improved that column by about 0.053 and then by about 0.025. For an isoform whose blind set differs from the labelled population, the holdout is a sample of the wrong population, and its sign cannot be trusted for changes aimed at that difference.

Limitations

At the latest reading, rank 41 puts this entry outside the top ten percent. Entries above it run six base model families against this one's three, pretrain every backbone on public bioactivity at scale, and in one case train on tens of thousands of unlabelled catalogue molecules chosen to resemble the blind set. None of those is a trick; they are infrastructure accumulated over a longer period.

The findings above are measured on one challenge with 4,905 training compounds and one blind set of 750, and the ones that were confirmed on blind data are confirmed by a single submission

each.

Reproduction

Code and notebooks are in this repository; the challenge data are not redistributed. The supporting record, including every closed direction with its measurement, is in `METHOD.md` and the notebooks.